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# 1 Preparation

## 1.1 Theoretical Background

### 1.1.1 Resonance absorption

When an atom in the excited state with Energy $E_a$ and average lifetime $\tau$ relaxes into the ground state with energy $E_g$ it emits a photon with the energy

$$E_\gamma = E_a - E_g - \frac{p_\gamma^2}{2M} = E_0 - E_R, \tag{1.1}$$

where $E_R = \frac{p_\gamma^2}{2M}$ is the energy resulting from the recoil the nucleus experiences due to the emission expressed in terms of the photon momentum $p_\gamma$ and the nucleus mass $M$. Similarly, a photon can also excite an atom from its ground state $E_g$ to the energy $E_a$.
In a quantum framework the energy $E_R$ has some uncertainty and the natural linewidth $\Gamma_0$ of the transition is then given by

$$\Gamma_0 \cdot \tau = \hbar. \tag{1.2}$$

The phenomenon of resonance absorption occurs when a photon emitted by one atom is then absorbed by another. Assuming both atoms are resting, this is only possible if $\Gamma_0$ is larger than the shift in energy due to the recoil. When dealing with optical transitions in nuclear shells, the relative linewidth $\Delta\nu/\nu$ is larger than $E_R/E_0$, the opposite being the case for $\gamma$-transitions in the nucleus.
In fig. 1.1 the broadening of the linewidth and the energy shift for emission and absorption respectively can be seen. Whereas it was previously assumed that our atoms were at rest, one has to take into account thermal movements of the respective atoms. In reality $E_\gamma$ becomes

$$E_\gamma = E_a - E_e - \frac{p_\gamma^2}{2M} + \frac{p_\gamma P}{M}\cos(\theta) \tag{1.3}$$

where $P$ is the momentum of the excited nucleus and $\theta$ is the angle between the nucleus and the emitted photon. Now, $\gamma$-transitions can be considered possible, although impropable, as the Energy spectrum, which experiences additional broadening due to the Doppler effect, is broad enough. When considering solids or molecules instead of free atoms one gets a correction term $-\Delta E_M$ in eq. (1.3) that accounts for the transitions of rotational and vibrational energy states in the lattice structure.

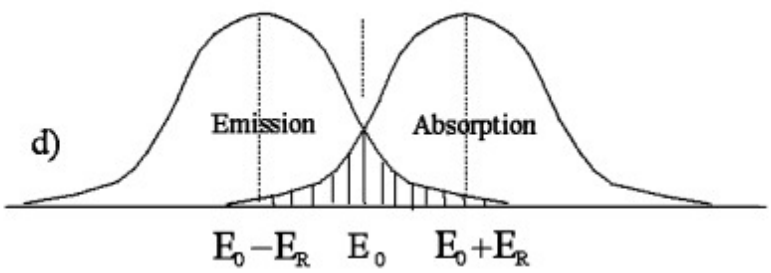


Figure 1.1: An illustration of the energy shift and line width of the $\gamma$ spectrum, extracted from [5, Chapter 7.12].

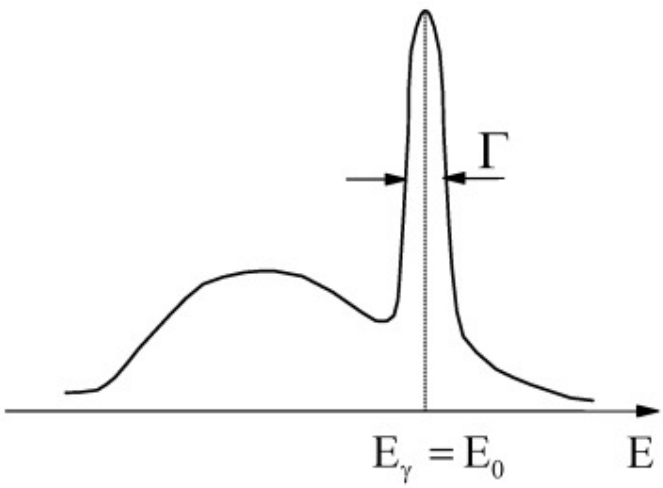


Figure 1.2: Schematic visualisation of a spectrum with one Moessbauer line, extracted from [5, Chapter 7.12].

### 1.1.2 The Mössbauer effect

In an ideal crystal lattice the recoil a free emitting nucleus would experience is distributed over the whole crystal, assuming that tight binding holds true. Small oscillations of the nuclei are possible however, as in reality the binding energy is finite. The resulting lattice vibrations are characterised by the creation and annihilation of phonons. If no phonons are created/annihilated this corresponds to a recoil-free emission. This recoil-free emission and absorption of $\gamma$-quanta is called the **Mössbauer effect** and appears in the resonance spectrum as a narrow and tall line, as depicted in fig. 1.2.

In this experiment, the Mössbauer effect of $^{57}$Fe will be analysed.

### 1.1.3 Experimental detection of the Mössbauer effect

Using conventional semiconductor detectors, the 14.4 keV cannot be detected because its linewidth $\Gamma_0$ is too narrow ($\Gamma_0 \sim 10^{-13}$ eV). However, through resonance absorption detection is possible. To achieve this, the emitted $\gamma$-quanta pass through an absorber and the intensity of the absoprtion is then measured. The respective energy is dependent on the overlap of apsorption and emission, which can be seen in fig. 1.1. By moving the emitter with a constant velocity $v$ one can manipulate the energy of $\gamma$-quanta. Mathematically, this results the Doppler effect results in an energy shift of

$$\Delta E_\gamma = \pm \frac{v}{c} \cdot E, \tag{1.4}$$

where $c$ is the speed of light and is chosen positive (+) when the emitter moves towards the absorber and negative (−) when it moves away from the absorber. This is necessary to compensate for shifts or splittings of Mössbauer line

### 1.1.4 Changes of the Mössbauer line

The effects which change the Mössbauer are line are hyperfine structure splitting, quadrupole splitting and the Isomeric shift. The **hyperfine structure splitting** is due to interaction with the internal magnetic field of the nucleus known as the Zeeman effect. Accordingly, the splitting in the energy is

$$\Delta E_m = -\frac{m}{I} \mu \cdot B \tag{1.5}$$

where $I$ is the nuclear spin and $m$ takes values from $-I, -I+1, 0, \ldots, I$. For resonace absorption the energy becomes

$$E_\gamma = -\frac{m_a}{I_a}\mu_a \cdot B + \frac{m_g}{I_g}\mu_g \cdot B \tag{1.6}$$

where $\Delta m = m_a - m_g$ is reduced to the possible values $0, \pm 1$ due to the selection rules for dipole transmission.
By introducing $E_0$ to account for the possible isomeric shift, setting $E_\gamma = E_0(1 + \frac{v}{c}$ and then solving for $v$ one obtains

$$v = \frac{(E_0' - E_0)c}{E_0} - \frac{m_a c}{I_a E_0}\mu_a \cdot B + \frac{m_g c}{I_g E_0}\mu_g \tag{1.7}$$

For $^{57}$Fe $I_a = 3/2$ and $I_g = 1/2$ so that six different splits of $v$, each with one resonance peak, are possible. The respective velocities are obtained by $v(m_a,\, m_g) = v^*(m_a,\, m_g) - I$ as $\sum_{m_a,\, m_g} = v^*(m_a,\, m_g) = 6I$ describes the isomeric shift. The resulting three positive velocities

$$v_i = \frac{1}{2}(v(m_a,\, m_g) - v(-m_a,\, -m_g)) \tag{1.8}$$

allow for different cases depending on the properties of the variables $A$ and $G$, where

$$A = \frac{\mu_a B c}{I_a E_0};\; G = \frac{\mu_g B c}{I_g E_0}, \tag{1.9}$$

namely, for $A > 0$, $G < 0$, $|A| > |G|$:

$$\begin{aligned} A &= v_1 - v_2, \\ G &= -v_2 + v_3, \\ 0 &= v_1 - 2v_2 + v_3. \end{aligned} \tag{1.10}$$

For $A > 0$, $G < 0$, $|A| < |G|$:

$$\begin{aligned} A &= v_1 - v_2, \\ G &= -v_2 - v_3, \\ 0 &= v_1 - 2v_2 - v_3. \end{aligned} \tag{1.11}$$

For $A < 0$, $G < 0$, $|A| > |G|$:

$$\begin{aligned} A &= -v_1 - v_2, \\ G &= -v_2 + v_3, \\ 0 &= v_1 - v_2 - 2v_3. \end{aligned} \tag{1.12}$$

For $A < 0$, $G < 0$, $|G/3| < |A| < |G|$:

$$\begin{aligned} A &= -v_1 - v_2, \\ G &= -v_2 + v_3, \\ 0 &= v_1 - v_2 - 2v_3. \end{aligned} \tag{1.13}$$

Which conditions are the case can be determined by measuring the respective velocities, so that one is able to then determine the magnetic field and moment.
Whether **quadrupole splitting** occurs depends on the relative position of the quadrupole due an asymmetric charge distribution of the nuclear spin $I$ to the electric gradient field. The energy splitting is:

$$\Delta E_Q(m) = \frac{eQ}{4}\left(\frac{\partial^2 V}{\partial z^2}\right)\frac{3m^2 - I(I+1)}{3I^2 - I(I+1)} = \pm\frac{eQ}{4}\left(\frac{\partial^2 V}{\partial z^2}\right), \tag{1.14}$$

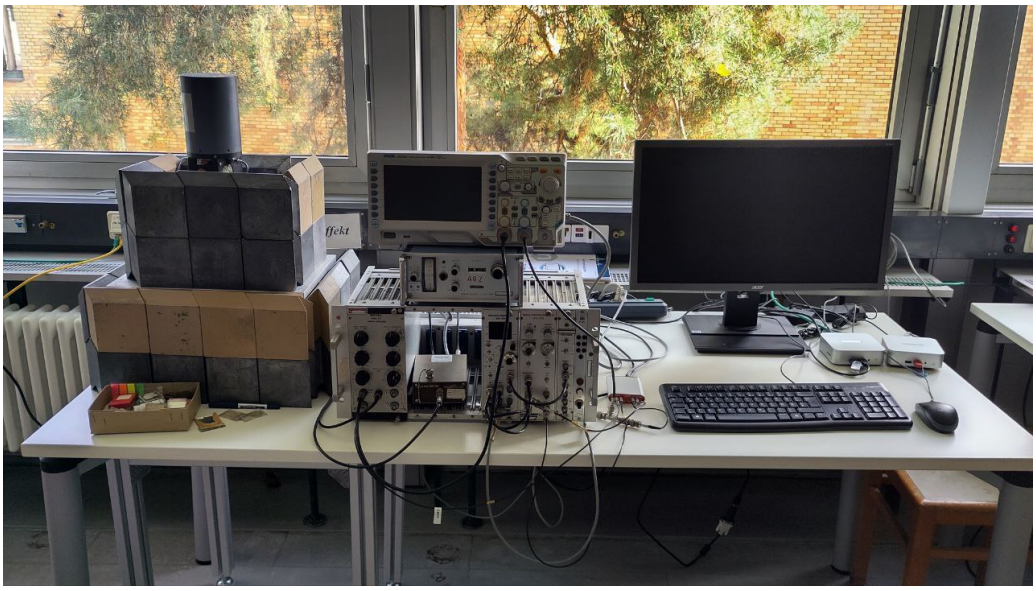

Figure 1.3: A picture of the working place with the experimental setup

as $m = m_g = \pm\frac{1}{2}, \pm\frac{3}{2}$, because quadrupole splitting can only for $I = I_g$ in our case. It is assumed that the electric gradient field points along the x-axis.
Thus we obtain for the velocity

$$v_Q = v_q(\pm 3/2) - v_Q(\pm 1/2) = \frac{ecQ}{2E_0}\frac{\partial^2 V}{\partial z^2}, \tag{1.15}$$

where $v_q(\pm 3/2)$ and $v_Q(\pm 1/2)$ correspond to the resulting two lines in the velocity spectrum.
Lastly, the **Isomeric shift** leads to another correction term for the transition energy. The isomeric shift results from the difference in the electron charge density between the emitter and the absorber atoms. This difference in charge density leads to a difference in the respective shift in the energy of the nuclear state caused by Coulomb repulsion:

$$\Delta E(R) = \frac{2\pi}{3}|\Psi(0)|^2 Ze\overline{R}. \tag{1.16}$$

Here, $\overline{R}$ is the nuclear charge radius, $Z$ is the charge number and $|\Psi(0)|^2$ is the electron charge distribution at the center of the nucleus.
The resulting Isomeric shift in the energy then is

$$E_\gamma = E_0 + \Delta E(R_a) - \Delta E(R_g). \tag{1.17}$$

## 1.2 Experimental Setup

The goal of this experiment is to measure the transmission rate of gamma rays through different absorbers. The experimental setup uses a radioactive $^{57}$Co source attached to a moving cylinder, while the absorbers are placed between the source and a Geiger counter detector. The $^{57}$Co operates as a single-line emitter and is diffused into Rhodium Rh, a material without an inner magnetic field. When an $^{57}$Co atom decays into an excited $^{57}$Fe nucleus, via electron capture, which decays further into its unexcited ground state. During the last process, a photon with an energy of precisely 14.4 keV is emitted, as explained in [5, Section 7.12]. This photon can be detected by the Geiger counter. Figure 1.3 shows a photo of the working placei[1]. An illustration of the experimental setup is shown in fig. 1.4, which was taken from [2].

[1]This photo was taken a couple of days before the experiment is performed.

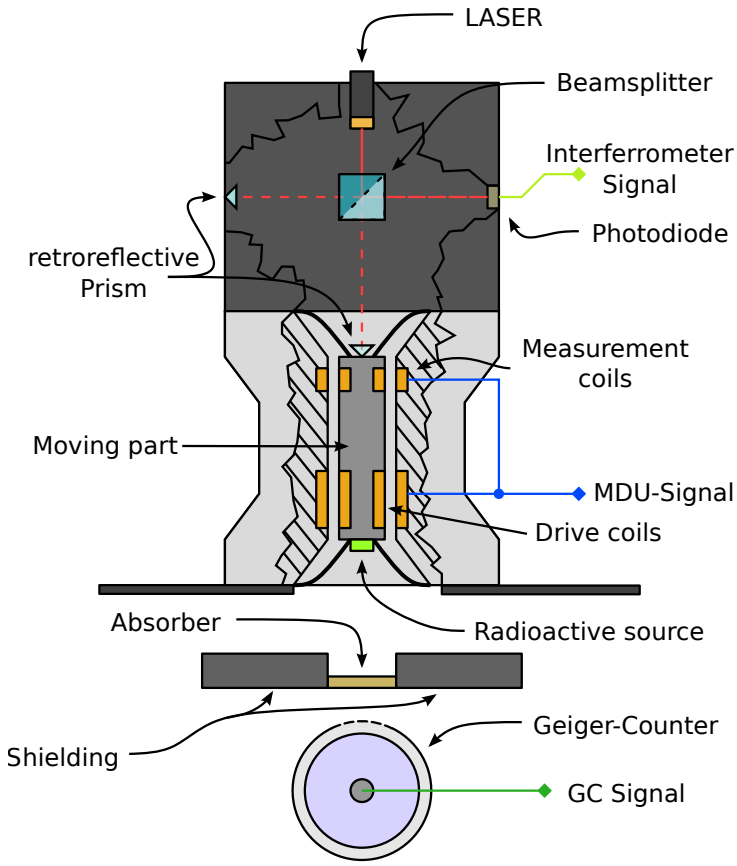


Figure 1.4: An illustration of the experimental setup for the Moessbauer Experiment

Figure 1.4 shows the moving cylinder, which operates similarly to a loudspeaker. Its speed is controlled by an electrical circuit, the Mössbauer Driving Unit (MDU), which receives a triangular analog voltage signal from a function generator, defining the velocity of the source. The cylinder is driven by a permanent magnet inside an electrical coil, which receives its voltage from the function generator signal. A second magnet and a second coil provide feedback to the MDU to stabilize the motion of the cylinder. The cylinder oscillates with a maximum velocity of $\pm 12\,\frac{\mathrm{mm}}{\mathrm{s}}$. An optimal movement is achieved when the frequency of the function generator matches the resonance frequency of the cylinder, typically between 17 Hz and 18 Hz according to [5, ]. With a positive voltage from the function generator, the source moves toward the absorber, increasing the $\gamma$-ray energy, whereas a negative voltage decreases the $\gamma$-ray energy.

### 1.2.1 Data Acquisition

Data acquisition is performed using an Multi Channel Pulse Height Analyser (MCA) implemented on a CAEN DT1260 digitizer, which is described in detail in [1].
The MCA is based on a 12 bit Analog to Digital Converter (ADC) that measures electrical voltages at a sampling rate of 65 Mega Samples per second (MS/s). A built-in FPGA performs real-time analysis of the incoming raw ADC signals as they are acquired. One of the two input channels is used to measure the velocity of the moving cylinder from a photodiode inside an interferomter, while the other detects electrical pulses from the Geiger counter, triggered by incident $\gamma$ photons.
The data is stored in a histogram, where each bin on the x-axis corresponds to a velocity of the cylinder and contains the number of registered events at the Geiger counter. The equal width of each bin corresponds to the duration for which a given velocity is applied during one full

oscillation period of the moving cylinder. In this setup, the histogram consists of $2^{10} = 1024$ bins of equal width, with each bin capable of storing up to 14 bit of event counts.

### 1.2.2 Velocity Measurement

To measure the actual velocity of the cylinder, a Michelson interferometer is attached. The interferometer splits a light beam from a laser with a wavelength of 650 nm into two orthogonal beams. One of the beams gets reflected from a Cat's eye prism at a fixed position, while the prism for the other beam is attached at the movable cylinder opposite to the $^{57}$Co source. The two reflected beams are recombined at the same beam splitter and directed to a photodiode, where the intensity of the interference signal, resulting from the phase shift between the two beams, is measured as an analog voltage.
When both prisms are stationary, the interference pattern, and thus the voltage measured at the photodiode, remain constant. Once the cylinder starts to move, the phase difference between the two beams becomes time dependent, causing the interference pattern to change. At the photodiode, alternating maxima and minima in intensity can be measured. The alteration frequency of these oscillations is proportional to the velocity of the moving cylinder.
The incoming beams can be described as two waves with a phase factor:

$$\begin{aligned} y_1 &= y_0 \cdot \sin\left(kx - \omega t\right), \\ y_2 &= y_0 \cdot \sin\left(kx - \omega t - \delta\right), \end{aligned}$$

where the phase factor $\delta$ results from the path length difference between the two beams:

$$\delta = k \cdot (L_1 - L_2) = k \cdot \Delta L = \frac{2\pi}{\lambda} \cdot \Delta L\,.$$

The intensity of the interfering beams is proportional to the square of the sum of both wave amplitudes. Assuming that the velocity $v$ is approximately constant over a short time interval $t$, we can express the path length difference as $\Delta L = 2vt$. Therefore, the intensity $S$ is proportional to

$$S = |y_1 + y_2|^2 \approx y_0^2 \cdot \cos^2\left(\frac{\delta}{2}\right) \approx \cos^2\left(\frac{2\pi vt}{\lambda}\right)\,. \tag{1.18}$$

Since the square of the cosine function has two maxima within a period, the frequency of these maxima can be related to the velocity of the moving cylinder. This frequency can be determined from the number of interference cycles $N$ measured at the photo diode during the measurement time $T$, corresponding to the duration in which a histogram channel is active and counts pulses:

$$f = 2 \cdot \frac{v}{\lambda} = \frac{N}{T}\,.$$

This relation allows the velocity to be expressed in terms of the number of recorded interference cycles:

$$v = \frac{\lambda}{2} \cdot \frac{N}{T} = \frac{N}{T} \cdot 325\,\mathrm{mm\,s^{-1}}\,. \tag{1.19}$$

The data acquisition device used in this experiment continuously calculates the velocity of the cylinder and uses it as an index for the corresponding bin to which the events detected by the Geiger counter are added. After sufficient measurement time, a reduced number of events will appear at specific velocities, indicating resonance absorption. This occurs when the additional Doppler energy resulting from the motion of the cylinder at the moment of decay of a $^{57}$Fe nucleus exactly compensates for the recoil energy $E_R$ of the nucleus, allowing the $\gamma$ photon to be absorbed by the material.

# 2 Analysis

## 2.1 Spectrum of the $^{57}$Co Source

At first, the energy spectrum of the $^{57}$Co source was recorded wit a MCA. It is depicted in fig. 2.1 with a clearly visible peak at channel 204, which is expected to be at 14.4 keV. The plot also shows a clearly visible noise at the left end of the spectrum. The signal to noise ratio is dependant on the activity of the source, which is decreasing with time because the source has a half life of 271.811(32) d, according to [6].

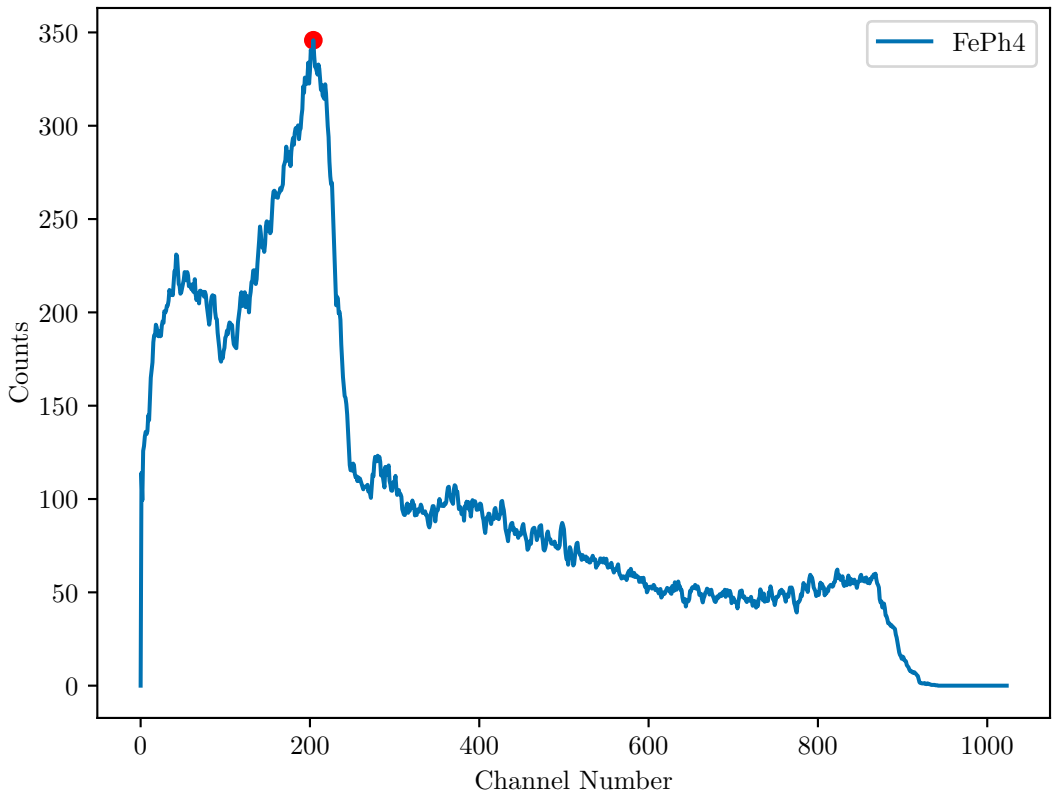


Figure 2.1: Energy Spectrum of the $^{57}$Co Source.

## 2.2 Velocity Calibration

To determine the velocity associated with each channel, a calibration run was performed. This involved counting the number of interference pattern repetitions during the time each velocity channel was active. Based on eq. (1.19), this count is used to calculate the actual velocity of the cylinder's motion within each velocity channel.
The total measurement time was 10 s, and with a fixed number of 1024 channels, the active time for each individual channel was given by:

$$T = \frac{10\,\mathrm{s}}{1024}$$

This equation was used to compute the velocity for each channel based on the number of measured interference pattern repetitions, as shown in fig. 2.2. Note that eq. (1.19) yields only the signed velocity, since the intensity function in eq. (1.18), based on $\cos^2$, is always positive. Consequently, velocities corresponding to the cylinder moving away from the detector were inverted. Both representations can be seen in fig. 2.2.

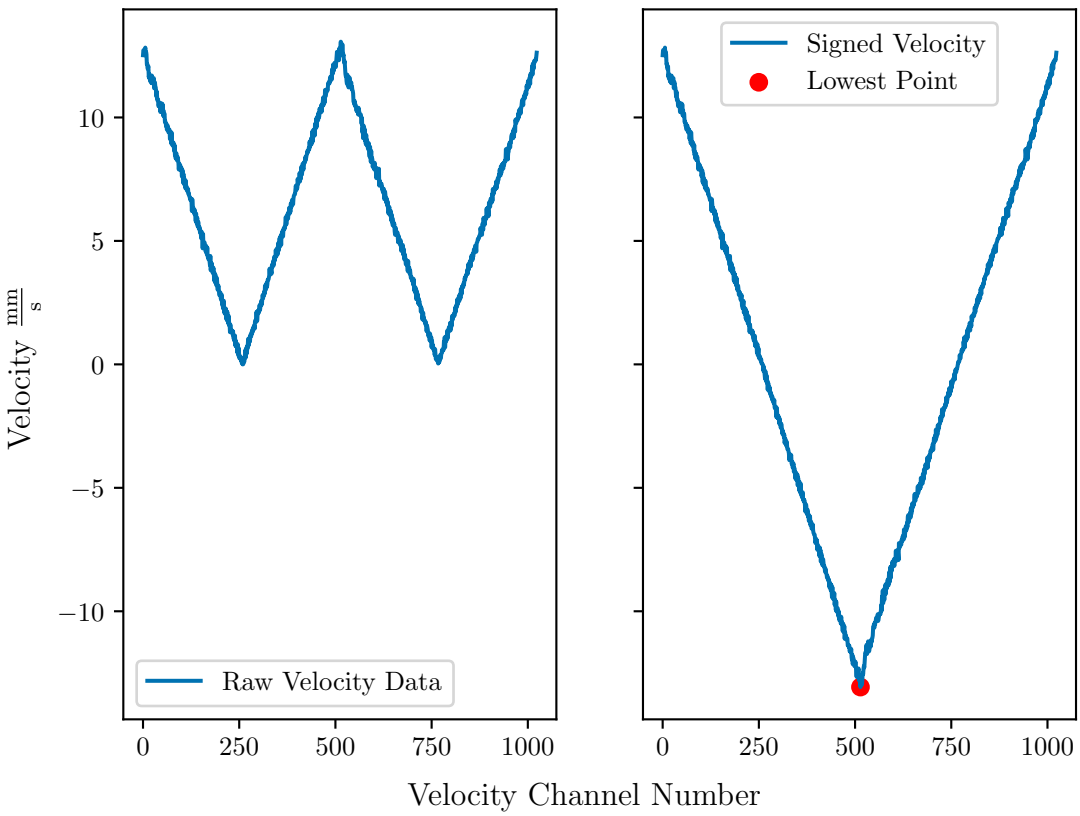


Figure 2.2: Raw velocity data, shown without (left) and with (right) sign correction.

To enable further use of the velocity calibration, a linear function was fitted to the data. Since the velocity signal transitions from its maximum to minimum and then back to maximum, each velocity channel appears twice in the dataset. For subsequent analysis, channels with matching velocities were combined.
Instead of splitting the dataset at the midpoint (channel 512), the turning point, where the velocity reaches its minimum at channel number 514, was chosen. The difference between the expected midpoint (512) and the actual turning point (514) was used to estimate the uncertainty in the channel index.
Having two measurements for each velocity value allowed for an estimation of the velocity

uncertainty. To align the lengths of both datasets, the initial samples were discarded. This has no impact on the further analysis, as these channels are not within the regions of interest.

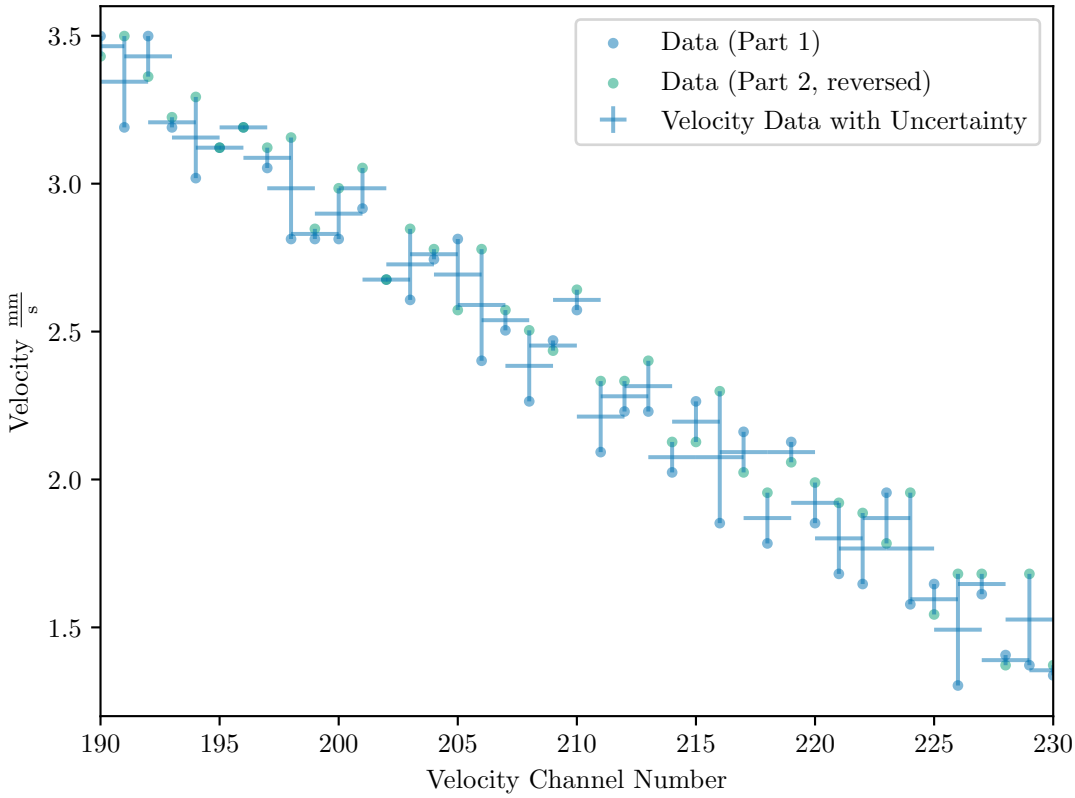


Figure 2.3: Uncertainty estimation for the Velocity Channels

Figure fig. 2.3 shows a cutout of the plot comparing the two velocity values per channel. The mean of both values was taken as the nominal velocity, and the uncertainty was defined to reach both data points.
The fits result is shown in fig. 2.4. The velocity for each channel can now be computed using the following linear equation, derived from the fit, where $N$ denotes the channel number from 0 to 512:

$$v(N) = -0.049\,416(28) \cdot N + 12.728(7) \tag{2.1}$$

The fit was performed using the `kafe2` package [3], taking into account the estimated uncertainties.

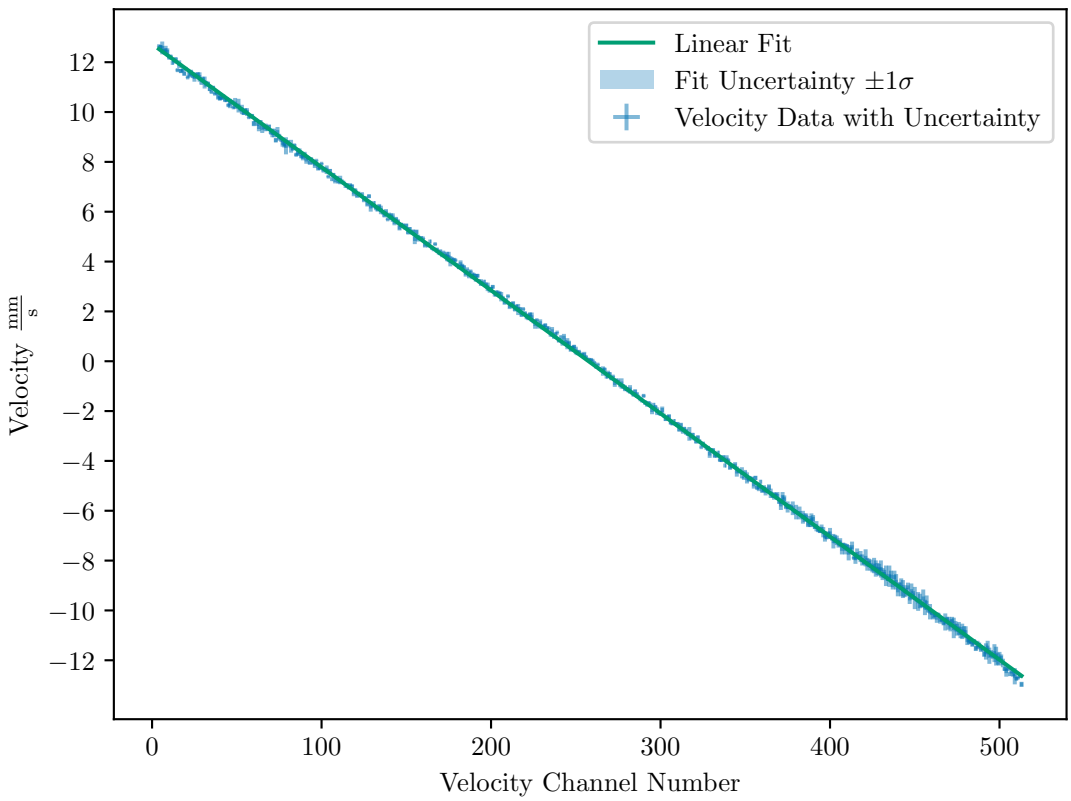


Figure 2.4: Velocity calibration: mapping velocity to each channel based on the number of recorded interference repetitions.

## 2.3 Stainless Steel

The resonance spectrum of Stainless Steel with one clearly visible resonance absorption peak can be seen in fig. 2.5. The spectrum which originally contained the velocity channelys from 0 to 1023, where every velocity was represented twice, was combined at the previously discovered velocity minimum at channel number 514 and the data for each channel was combined.
Since the recorded counts in every channel are poisson distributed, an unvertainty of $\sqrt{N}$ was assumed for each individual channel.
For the analysis, a combination of two Breit-Winger functions was fitted to the data, which can be seed in fig. 2.5. The peak of the fit function was located at

$$E_0 = -10.4(9)\,\mathrm{neV}\,. \tag{2.2}$$

The resulting peak of the fit had an uncertainty of

$$\Gamma = 19.4(24)\,\mathrm{neV} = 3.1(4) \times 10^{-18}\,\mathrm{nJ}\,. \tag{2.3}$$

From this value, the lifetime of the excited state is calculated:

$$\tau = \frac{\hbar}{\Gamma} = 1.69(21) \times 10^{-8}\,\mathrm{s}\,. \tag{2.4}$$

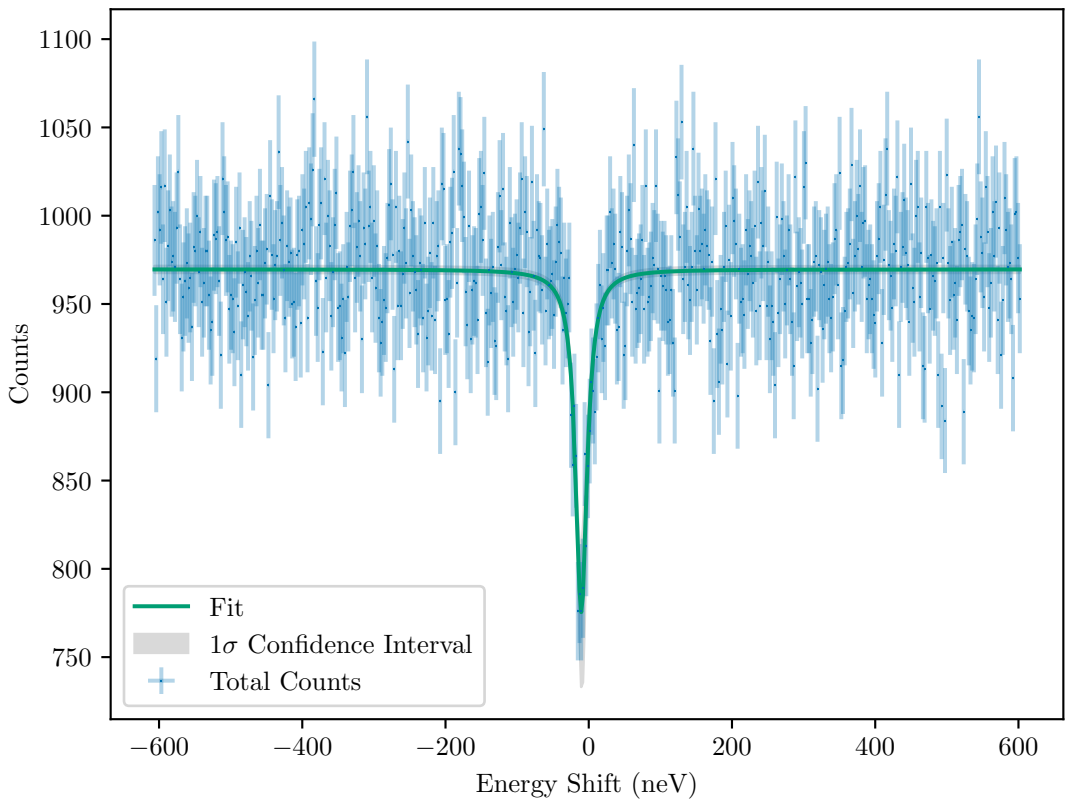


Figure 2.5: Spectrum and Breit-Wiger fit for Stainless Steel

## 2.4 Natural Iron

The resonance spectrum of Natural Iron can be seen in fig. 2.6. The spectrum which originally contained the velocity channelys from 0 to 1023, where every velocity was represented twice, was combined at the previously discovered velocity minimum at channel number 514 and the data for each channel was combined.
The peaks are find at the following velocities:

- $-5.469(11)\,\mathrm{mm\,s^{-1}}$
- $-3.179(10)\,\mathrm{mm\,s^{-1}}$
- $-0.875(13)\,\mathrm{mm\,s^{-1}}$
- $0.813(11)\,\mathrm{mm\,s^{-1}}$
- $3.180(11)\,\mathrm{mm\,s^{-1}}$
- $5.529(12)\,\mathrm{mm\,s^{-1}}$

The magnetic moment of the excited state results in

$$\mu_a = -0.1530(18)\mu_N \tag{2.5}$$

where $\mu_n$ refers to the nuclear magneton. For the magnetic moment at the ground state $\mu_g$ the value of $\mu_g = 90.3(7) \times 10^{-3}\,\mu_N$ is used and the spins are $I_g = \frac{1}{2}$ and $I_a = \frac{3}{2}$. The magnetic field at the nucleus is:

$$B = 34.20(29)\,\mathrm{T}\,. \tag{2.6}$$

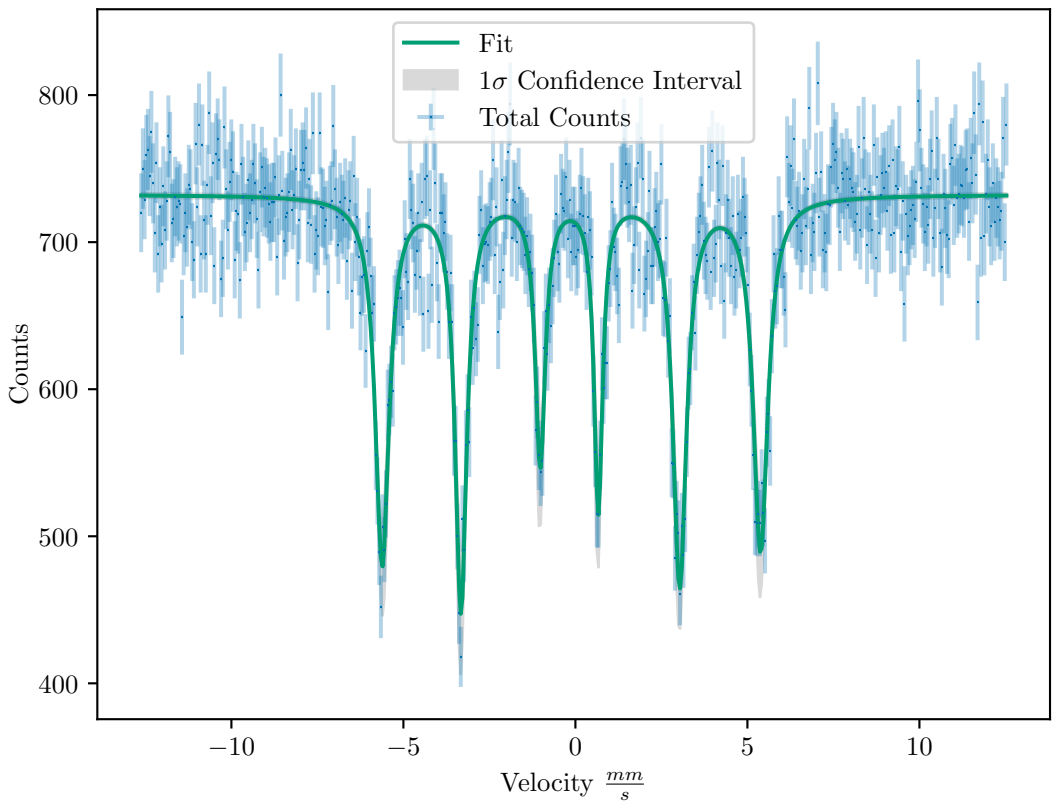


Figure 2.6: Spectrum and Multi-Breit-Wiger fit for Natural Iron

## 2.5 Iron Phosphate $FePo_4$

The resonance spectrum of $FePo_4$ can be seen in fig. 2.7. The spectrum which originally contained the velocity channelys from 0 to 1023, where every velocity was represented twice, was combined at the previously discovered velocity minimum at channel number 514 and the data for each channel was combined.
For the analysis, a combination of two Breit-Winger functions was fitted to the data, which can be seed in fig. 2.7. Unfortunately, the data points are not clearly distinguishable from each other beacuse the two absorption regions are very close to each other and the low event rate of the $^{57}$Co source made the signal to noise ratio very bad. Trying to move the initial values for the fit closer together created the problem that the two regions were registerd as a single region with a wide width. The resulting energy shifts at which an absorption can be discovered are:

$$E_0 = -6.5(12)\,\text{neV}$$
$$E_1 = 31.8(20)\,\text{neV}\,.$$

Fro these values, the resulting isomeric shift is:

$$I = \frac{E_0 + E_1}{2} = 12.6(11)\,\text{neV}\,. \tag{2.7}$$

The two resonance absorption peaks are comming from the Quadrupole splitting. With the known quadrupole moment $Q = 0.21(1) \times 10^{-28}\,\text{m}^2$ the second derivative of the electric potential at the nucleus can be determined:

$$\partial_z^2 V = \frac{2}{Q}(E_1 - E_0) = 3.64(28) \times 10^{21}\,\text{V}\,\text{m}^{-2}\,. \tag{2.8}$$

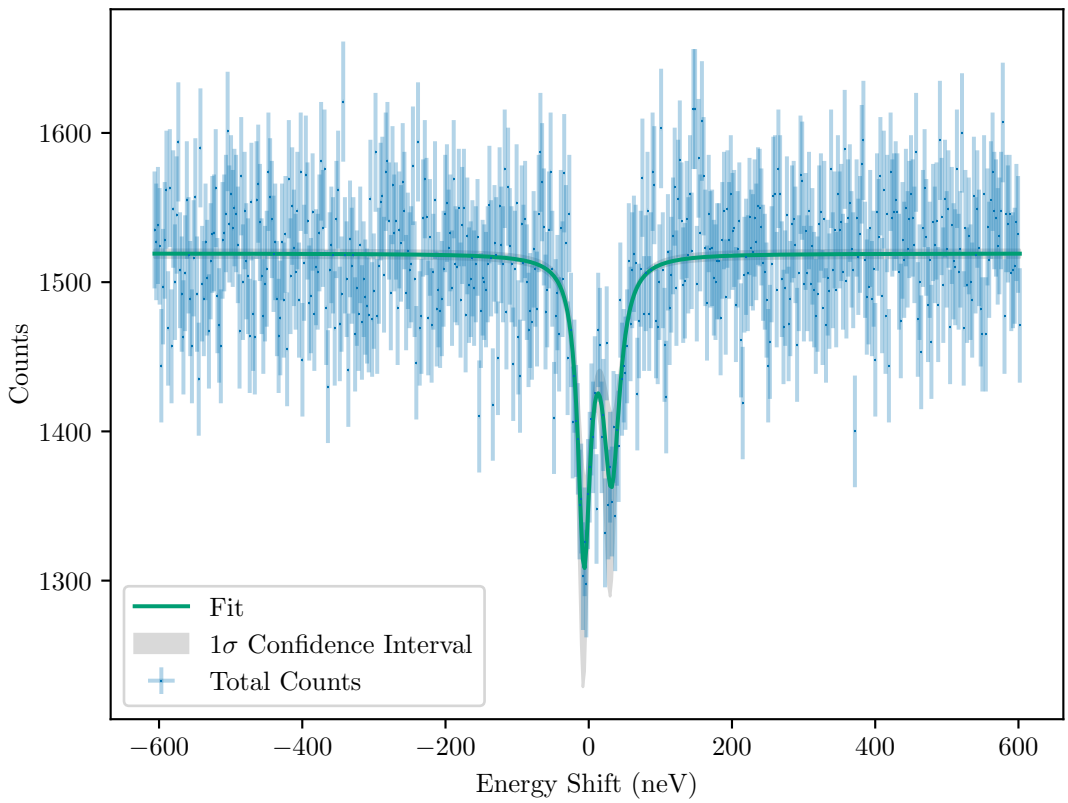


Figure 2.7: Spectrum and double Breit-Wiger fit for $FePo_4$

## 2.6 Iron Sulphate $FeSo_4$

For the resonance spectrum of $FeSo_4$, which can be seen in fig. 2.8, the performed analysis is the sams as for the previous spectrum. The enregy Shifts from the fit are:

$$E_0 = -13.6(10)\,\mathrm{neV}$$
$$E_1 = 124.6(12)\,\mathrm{neV}\,.$$

and the derivative of the electric potential at the nucleus is:

$$\partial_z^2 V = 1.32(6) \times 10^{22}\,\mathrm{V\,m^{-2}}\,. \tag{2.9}$$

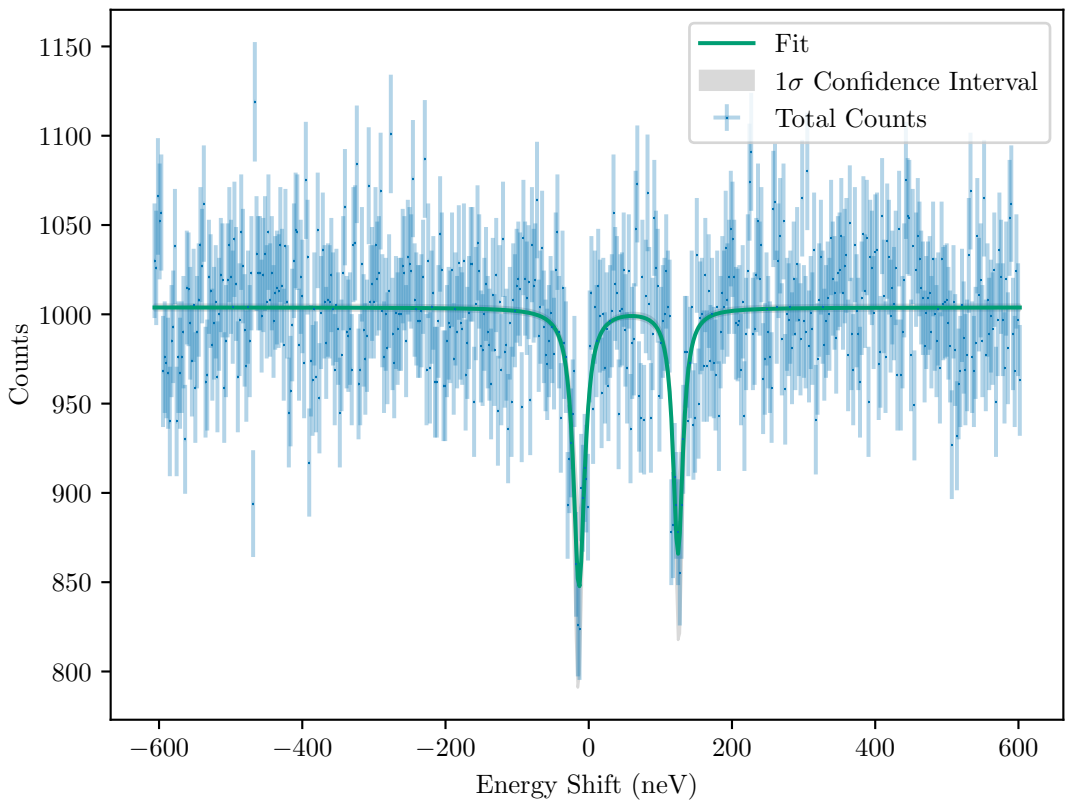


Figure 2.8: Spectrum and double Breit-Wiger fit for $FeSo_4$

# 3 Discussion

The literature values are taken from the paper "Development of a Mössbauer Spectrometer" [4].

## 3.1 Results in comparison to the literature values

### 3.1.1 Stainless Steel

The result obtained for $E_0$ eq. (2.2) corresponds directly to the Isomeric shift $I_0$. From the literature, we know that $I_{0(\text{lit})} = -12.0(5)\,\text{neV}$. The literature values is then within the $2\sigma$ range of our result $E_0 = -10.4(9)\,\text{neV}$. The literature value $\Gamma_{\text{lit}} = -10.5(4)\,\text{neV}$, however, is only within $37\sigma$ range of our result $\Gamma = -19.4(24)\,\text{neV}$. Accordingly, there is a strong deviation of the calculated lifetime from the expected value.

### 3.1.2 Natural Iron

For the calculation of $B$ and $\mu_a$ the case $v_1 - 2v_2 + v_3 = 0$ was chosen as the result from the previously determined velocities was $-0.04(98)\,\text{neV}$ and thus within the $1\sigma$ range of the theoretical result. For $B$ and $\mu_a$ respectively the literature values are $B = 32.4(1)\,\text{T}$ and $\mu_{a,1} = -157(3) \cdot 10^{-3}\mu_N$ and $\mu_{a,2} = -154.9(2) \cdot 10^{-3}\mu_N$. The literature value for $B$ our result is in the $5\sigma$ range of our result and therefore imprecise, whereas the literature values for $\mu_a$ are within the $3\sigma$ and the $2\sigma$ interval of our result respectively.

### 3.1.3 Iron Phosphate/Iron Sulfate

As there were no literature values available for iron phosphate and iron sulfate a discussion of the results for either of them is not possible.

## 3.2 Precision and Errors

Two things can be observed in our comparison of our results with the literature values: First, we have results that are extremely close to the literature value and second, we have results that are quite different than the literature values. In the latter case, we have $\Gamma$ and $B$. For $\Gamma$ we must attribute the difference to the age and thus the reduced activity of our source. For $B$ we can only speculate that our energy value may be the cause of the difference, as we have no literature value for the energy.

In the former case, we have $I_0$ and $\mu_a$. Here, the precision of both results demonstrates the accuracy of our measurements. This accuracy is also evident by the comparitively small errors, which in turn were used for meticulous error calculation, where we also used the uncertainties from the calibration for error propagation.

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